MIL-C-24643/51B
3 October 1986
SUPERSEDING
MIL-C-0024643/51A(SH)
8 November 1985
MIL-C-24643/51
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, MULTICONDUCTOR, TYPE LSMNW
(INCLUDING VARIATION LSMNWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LSMNW (unarmored) and LSMNWA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated. AWG size 18, class B stranded.
Second - Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene or ethylene propylene rubber insulation, colored black, 0.020 inch nominal thickness. Standard identification code applied by method 1.
Fourth - The required number of conductors (see table I), cabled together with a lay not greater than 24 times the pitch diameter of the layer, except that the lay of the two outer layers for size 44 shall not be greater than 18 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be employed, as required, to form a firm, well-rounded assembly.
Fifth - Binder tape applied helically with overlap.

(B) denotes changes.

AMSC N/A FSC 6145
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Construction variant for type LSMNW

Sixth - Cross-linked polyolefin jacket. Cable jacket thickness 0.050 inch nominal. Cable surface marking required.

Construction variant for type LSMNWA

Sixth - Cross-linked polyolefin jacket. Cable jacket thickness 0.050 inch nominal.
Seventh - Braided metal armor and paint.

TABLE I. Details.

Military 1/ part no. M24643/51	Size 2/	Number of conductors	Overall diameter (inches)				Cold bending mandrel diameter maximum (inches)
			Type LSMNW		Type LSMNWA		
			minimum	maximum	minimum	maximum	
-01UN	7	7	0.370	0.400	0.420	0.450	3
-02UN	10	10	.457	.495	.507	.545	3
-03UN	14	14	.494	.535	.544	.585	4
-04UN	19	19	.545	.590	.595	.640	4
-05UN	24	24	.633	.685	.683	.735	5
-06UN	30	30	.670	.725	.720	.775	5
-07UN	37	37	.726	.785	.776	.835	6
-08UN	44	44	.823	.890	.873	.940	7

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LSMNW and LSMNWA.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	7.07
Voltage withstand - volts, root mean square, minimum....	3000
Insulation resistance - megohms/1000 feet, minimum.....	1000
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group B:	
Cold bending, cable - (See table II for mandrel diameter)	No damage
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Cross-linked polyethylene	
Tensile strength - lbf/in^2, minimum	1800
Elongation - percent, minimum	250
Ethylene propylene rubber	
Tensile strength - lb/in, minimum	700
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lbf/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Ethylene-propylene rubber	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged, minimum	75
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (conductors) - cycles, minimum	25
Permanence of printing (jacket) type LSMNW only - cycles, minimum	250
Cable filler removability	No failure
Heat distortion - percent of unaged, maximum	30
Armor - conformance to material, construction and coverage, type LSMNWA only	No failure

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

	Requirements
Cold working (cable) (minus 20 ± 2°C)	No damage
Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	--
(B) Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSMNW-7	1500
LSMNW-10	1500
All other sizes	1000

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-1005-51)

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

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